Subsequential Infinite-Volume Limits of the Wilson Loop Functions

Abstract

The limits of the renormalised Wilson loop expectations constructed in the paper live on a four-dimensional torus, and the bound on which their existence rests grows exponentially with the volume of the torus. We show that a chessboard estimate, which uses nothing but the reflection positivity of the lattice measures, replaces this bound by one which does not depend on the period of the torus. As a consequence the limits converge along a subsequence of periods, for every ensemble of admissible loops in $\mathbb{R}^4$ at once, and the limiting functions are bounded, reflection positive, and covariant under the symmetries which the paper establishes on the torus. Their average over the permutations of the axes is, in addition, covariant under the whole hypercubic group, and it remains bounded and reflection positive. Finally, for two parts of an ensemble on the two sides of a coordinate hyperplane, the limits are locally Lipschitz functions of the distance by which the parts are moved apart along its normal, on each of the two classes of distances which the lattices distinguish.

1 The Result

Everything in the paper takes place on the torus $(\mathbb{R}/T\mathbb{Z})^4$ of the period $T = 2L^{m_T}u$, and we ask what happens as $m_T \to \infty$. Most of the construction does not see the period. The bare coupling $g_0(K, \bar{g})$ of Paper, Theorem 4.1, is determined by the coefficients β_{k+1} of Paper, (4.6), which are defined on the infinite lattice [1, (1.20)–(1.22), p. 264], so that the map from the bare coupling to the running couplings, and with it the bare coupling for a target $(K, \bar{g})$, may be chosen the same for all periods; we make this choice. The step factors, and therefore the normalisations, depend neither on the loop nor on the torus (Paper, Proposition 7.13), and the threshold γ_0 and the function $q_\sharp$ of Paper, Theorem 1.5, depend only on the constants of the construction and on Λ_*. Of the three quantities which enter the uniform bound of Paper, Theorem 8.6, the sum $B_E(\mathcal{C})$ of the remainders does not involve the torus, and the insertion total $I_*(\mathcal{C})$ decreases as the torus grows. The third one grows, however: the bound $\bar{\delta}_{\mathrm{LF}}$ on the weight of the large-field terms (Paper, Proposition 8.4) contains the factor e^{2C_K} with $C_K = C_{\mathrm{fs}}N_q^4(1 + \log g_{K'}^{-1})$, where N_q is the number of sites per side of the lattice at the stopping scale (Paper, (8.2)). This factor is the price of comparing whole integrals over that lattice, and it makes the bound of Paper, Theorem 8.6, exponential in the volume.

Reflection positivity turns a cost which is extensive in the volume into a cost per block. If the torus is cut into $|\Lambda|$ congruent blocks and an ensemble is placed in one of them, the chessboard estimate bounds the $|\Lambda|$-th power of its expectation by the expectation of the ensemble obtained by reflecting it into every block. The uniform bound of the paper, applied to this larger ensemble, contains the factor e^{2C_K} only once, so that its $|\Lambda|$-th root is a number of the block alone, while the costs of the loops, which are sums over the loops, are divided back to those of one copy. The chessboard estimate is due to Fröhlich, Israel, Lieb, and Simon

[3]; we prove the form which we need, from the reflection positivity of the lattice measures in Paper, Lemma 12.2, in Section 2.

Throughout, L, u, Λ_*, γ_0, and $q_\sharp$ are as in Paper, Theorem 1.5, and $\bar g \in (0,\gamma_0]$ is fixed. For an integer $s \ge 0$ we write $T_s := 2L^s u$ and $\mathbb{T}^{(s)} := (\mathbb{R}/T_s\mathbb{Z})^4$, which is the torus of the paper with $m_T = s$, and a superscript (s) marks the objects of the paper on it, such as $\mathbb{E}_K^{(s)}$, $W_K^{(s)}$, and $W^{(s)}$. The limit in the volume concerns loops in $\mathbb{R}^4$, which we now define.

Definition 1.1 (Ensembles in $\mathbb{R}^4$)**.** Let $q \ge 0$ be an integer. A *loop at level q in $\mathbb{R}^4$* is a closed path of bonds of the lattice $uL^{-q}\mathbb{Z}^4$ which visits no site twice, with its orientation, and a *loop ensemble at level q in $\mathbb{R}^4$* is a finite family $\mathcal{C} = ((C_1,\chi_1),\dots,(C_n,\chi_n))$ of pairwise vertex-disjoint loops at level q in $\mathbb{R}^4$, each with the character χ_i of an irreducible unitary representation of G. The *extent* $E(\mathcal{C})$ is the largest of the four numbers $\max x_\mu - \min x_\mu$, the maximum and the minimum being taken over the sites x of the loops of $\mathcal{C}$. We put $b_q(\mathcal{C}) := uL^{n_q(\mathcal{C})}$, where $n_q(\mathcal{C})$ is the least integer $n \ge -q$ with $uL^n \ge E(\mathcal{C}) + 2uL^{-q}$, and $s_q(\mathcal{C}) := \max(0, n_q(\mathcal{C}) + 1)$. Finally $\mathfrak{E}$ denotes the set of the pairs $(\mathcal{C},q)$ of an integer $q \ge 0$ and a loop ensemble $\mathcal{C}$ at level q in $\mathbb{R}^4$ whose loops are admissible in the sense of Paper, Definition 1.4, and for which $q \ge q_\sharp(c_{\mathcal{C}},\bar g)$.

Let $(\mathcal{C},q) \in \mathfrak{E}$ and $s \ge s_q(\mathcal{C})$, and let $\pi_s\colon \mathbb{R}^4 \to \mathbb{T}^{(s)}$ be the covering map. The sites of $\mathcal{C}$ lie in a closed box of side $E(\mathcal{C}) < b_q(\mathcal{C})$, and π_s is injective on every box of side less than $T_s = 2uL^s \ge 2Lb_q(\mathcal{C})$. The family $\pi_s\mathcal{C}$ of the images, with the same characters, is therefore a loop ensemble at level q on $\mathbb{T}^{(s)}$ in the sense of Paper, Definition 1.1, the condition $3\,\mathrm{diam}(\pi_s C_i) < T_s$ holding because $\mathrm{diam}(C_i) \le E(\mathcal{C}) < b_q(\mathcal{C})$; and its loops have the chord–arc constants, the ratios P/r, and the numbers P_q of the loops of $\mathcal{C}$, since these are computed in a lift. We write $W^{(s)}(\mathcal{C})$ for the limit of Paper, Theorem 1.5, of the ensemble $\pi_s\mathcal{C}$ with the normalisation of the level q.

Two numbers enter the bound. The first is the number $B_E(\mathcal{C})$ of Paper, (7.32), which is defined without reference to the torus and bounds the sum of the remainders on every torus. The second replaces the insertion total of Paper, (8.37), which involves the torus only through its last factor $S_{\mathrm{wr}}(c_{\mathrm{att}}N_q/M)$; since $S_{\mathrm{wr}}(b) = \sum_{i\ge0} L^i e^{-bL^i}$ decreases in b and $N_q = 2L^{m_T+j_q} \ge 2$, the insertion total is at most

$$I_*^\flat(\mathcal{C}) := P_{K'}(\mathcal{C})\Big[\frac{2c_\sigma C_\chi(1+2\kappa_s)B_3 S_w\,\bar\rho\,\varepsilon_{K'}}{\min(\alpha_{0,K'},\alpha_{1,K'})} + C^2\bar\rho\,S_{\mathrm{wr}}\Big(\frac{2c_{\mathrm{att}}}{M}\Big)\Big]\,, \tag{1.1}$$

with the constants of Paper, (8.37). For a length $b > 0$ we put, with the constants of Paper, Proposition 8.4, and with $\hat g_{K'}^{-2} = \bar g^{-2} + b' j_q + C_W$ (Paper, (3.3)),

$$\Gamma_q(b,\bar g) := \Big(\frac{bL^{j_q}}{u}\Big)^4\Big[2C_{\mathrm{fs}}\Big(1+\frac12\log\hat g_{K'}^{-2}\Big) + \frac{2}{(MR_{K'})^4}\exp(-c_{\mathrm{LF}}\,p_0((1+\beta_0)\bar g))\Big]\,. \tag{1.2}$$

The numbers $P_{K'}(\mathcal{C}) = P_q(\mathcal{C})L^3MR_{K'-1}$, $\varepsilon_{K'}$, $\alpha_{0,K'}$, $\alpha_{1,K'}$, $R_{K'}$, and $\hat g_{K'}$ are functions of q, $\bar g$, and the constants of the construction (Paper, Section 8), and the constants in (1.1) and (1.2) do not depend on the torus (Paper, Definition 5.1, uniformity, and (8.29)). Hence neither $I_*^\flat(\mathcal{C})$ nor $\Gamma_q(b,\bar g)$ depends on the period or on K; the first is a sum over the loops of $\mathcal{C}$, and so is $B_E(\mathcal{C})$.

The theorem says that the limits on the tori converge along one subsequence of periods for all ensembles at once, and that the limits keep the bound and the structure which the paper proves on each torus.

Theorem 1.2 (Subsequential Infinite-Volume Limits)**.** *There is a strictly increasing sequence $(s_k)_{k\geq 1}$ of integers such that, for every $(\mathcal{C}, q) \in \mathfrak{E}$, the limit*

$$W^{\infty}(\mathcal{C}) \coloneqq \lim_{k\to\infty} W^{(s_k)}(\mathcal{C}) \tag{1.3}$$

exists, the level being that of the pair. For every level q and all ensembles $\mathcal{C}$, $g\mathcal{C}$, $h\mathcal{C}$, $\mathcal{D}_a$ below which form pairs of $\mathfrak{E}$ with this level, the limits have the following properties.

(i) $|W^{\infty}(\mathcal{C})| \leq \exp(B_E(\mathcal{C}) + I^{\flat}_{*}(\mathcal{C}) + \Gamma_q(b_q(\mathcal{C}), \bar{g}))$.

(ii) $W^{\infty}(\mathcal{C})$ *does not depend on the order of the loops of* $\mathcal{C}$*;* $W^{\infty}(g\mathcal{C}) = W^{\infty}(\mathcal{C})$ *for every map* $g(x) = Ox + v$ *of* $\mathbb{R}^4$ *with a diagonal matrix* O *with entries* ± 1 *and* $v \in uL^{-q}\mathbb{Z}^4$*;* $W^{\infty}(\bar{\mathcal{C}}) = \overline{W^{\infty}(\mathcal{C})}$*, and* $W^{\infty}(\mathcal{C})$ *does not change if a loop* C_i *is replaced by* C_i^{-1} *and its character by* $\bar{\chi}_i$*; and* $W^{\infty}(h\mathcal{C}) = W^{\infty}(\mathcal{C})$ *for every map* $h(x) = Ox + v$ *with a signed permutation matrix* O *and* $v \in uL^{-q}\mathbb{Z}^4$ *for which there are maps* $g_1, \dots, g_n$ *of the previous kind with* $hC_i = g_iC_i$ *for every* i*.*

(iii) Let $\mu \in \{0, \dots, 3\}$ *and* $y \in uL^{-q}\mathbb{Z}$*, let* θ *be the reflection of* $\mathbb{R}^4$ *which replaces* x_μ *by* $2y - x_\mu$*, and let* $\mathcal{D}_1, \dots, \mathcal{D}_m$ *be loop ensembles at level* q *in* $\mathbb{R}^4$*, possibly empty, whose loops are admissible and lie in* $\{x_\mu > y\}$*, with* $q \geq q_\sharp(c, \bar{g})$ *for the largest number* c *of the form* c_π *among their representations. Then, with* $\mathcal{D}_0 \coloneqq \emptyset$ *and* $W^{\infty}(\emptyset) \coloneqq 1$*, the matrices*

$$(W^{\infty}(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b))_{1\leq a,b\leq m} \qquad \text{and} \qquad (W^{\infty}(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) - W^{\infty}(\mathcal{D}_a)\overline{W^{\infty}(\mathcal{D}_b)})_{1\leq a,b\leq m} \tag{1.4}$$

are Hermitian and positive semidefinite, and $W^{\infty}(\theta\bar{\mathcal{D}}_a) = \overline{W^{\infty}(\mathcal{D}_a)}$ *for every* a*.*

The bound (i) is the one which the compactness argument needs, a number of the ensemble which does not depend on the period. It is not a product over the loops, since it contains the fourth power of the extent of the whole ensemble through $b_q(\mathcal{C})$, and Remark 4.1 says what the theorem leaves open.

The covariance in (ii) under a map which exchanges two axes is restricted, although the lattice measures are invariant under every such map (Paper, Lemma 12.1): the normalisations are not known to be invariant under the exchange of two axes (Paper, Section 12.3). The full hypercubic group is recovered by averaging over the permutations of the axes. For a permutation σ of $\{0, \dots, 3\}$ let P_σ be the linear map of $\mathbb{R}^4$ with $(P_\sigma x)_{\sigma(\mu)} = x_\mu$, so that $P_\sigma e_\mu = e_{\sigma(\mu)}$ and $P_\sigma P_\tau = P_{\sigma\tau}$. It maps $uL^{-q}\mathbb{Z}^4$ onto itself and preserves the distance $|\cdot|_\infty$, the segments, and the corners, so that $P_\sigma\mathcal{C}$, the family of the images with the same characters, has the chord–arc constants, ratios P/r, numbers P_q, extent, and number c of $\mathcal{C}$, and $(P_\sigma\mathcal{C}, q) \in \mathfrak{E}$ whenever $(\mathcal{C}, q) \in \mathfrak{E}$. We put

$$W^{\mathrm{hc}}(\mathcal{C}) \coloneqq \frac{1}{24}\sum_{\sigma} W^{\infty}(P_\sigma\mathcal{C})\,. \tag{1.5}$$

On each torus $\mathbb{T}^{(s)}$ the map with the formula of P_σ is a symmetry of level q, so that $\mathbb{E}^{(s)}_K[W_{\pi_s P_\sigma \mathcal{C}}] = \mathbb{E}^{(s)}_K[W_{\pi_s\mathcal{C}}]$ by Paper, Lemma 12.1. Hence $W^{\mathrm{hc}}(\mathcal{C})$ is the limit, as $K \to \infty$ and then along (s_k), of $\mathbb{E}^{(s)}_K[W_{\pi_s\mathcal{C}}]$ multiplied by $\frac{1}{24}\sum_\sigma \prod_i \kappa^{(q)}_K(P_\sigma C_i, \chi_i)^{-1}$: the expectation divided by a normalisation which is invariant under the whole hypercubic group, but which is an average of products over the loops rather than a product. If the normalisations are invariant under the exchanges of axes, then $W^{\mathrm{hc}} = W^{\infty}$.

The translations in Theorem 1.2(ii) are restricted to the vectors of $uL^{-q}\mathbb{Z}^4$ only because the loops are paths in this lattice, whereas neither the tori nor the measures on them single

out an origin, and we remove the restriction by a convention. A *translate* of a loop ensemble $\mathcal{C}$ at level q in $\mathbb{R}^4$ is the family $\mathcal{C}+w$ of its loops shifted by one vector $w \in \mathbb{R}^4$, with the same characters, and for $(\mathcal{C}, q) \in \mathfrak{E}$ and $w \in \mathbb{R}^4$ we put

$$W^\infty(\mathcal{C}+w) \coloneqq W^\infty(\mathcal{C}) \,, \qquad W^{\mathrm{hc}}(\mathcal{C}+w) \coloneqq W^{\mathrm{hc}}(\mathcal{C}) \,, \tag{1.6}$$

the level being q. These numbers depend only on the family $\mathcal{C}+w$ and on q. Indeed, let $\mathcal{C}+w = \mathcal{C}'+w'$ with a second loop ensemble $\mathcal{C}'$ at level q, so that $\mathcal{C}' = \mathcal{C}+t$ with $t \coloneqq w-w'$. Since a loop is a closed path without self-intersections, it is not contained in a line and has a corner; the corners of $\mathcal{C}$ and of $\mathcal{C}'$ are sites of $uL^{-q}\mathbb{Z}^4$, and the translation by t maps the former to the latter, so that $t \in uL^{-q}\mathbb{Z}^4$. Theorem 1.2(ii) therefore gives $W^\infty(\mathcal{C}') = W^\infty(\mathcal{C})$ and, applied to $P_\sigma\mathcal{C}' = P_\sigma\mathcal{C} + P_\sigma t$ for every σ, $W^{\mathrm{hc}}(\mathcal{C}') = W^{\mathrm{hc}}(\mathcal{C})$. The convention is the result of carrying out the construction on the lattices $w + uL^{-K}\mathbb{Z}^4$ in place of $uL^{-K}\mathbb{Z}^4$, since the translation by w is an isometry of every torus $\mathbb{T}^{(s)}$ which carries the lattices, the measures, and the loops of the one construction to those of the other.

Corollary 1.3 (Hypercubic Covariance)**.** *For every level q and all ensembles $\mathcal{C}$, $\mathcal{D}_a$ below which form pairs of $\mathfrak{E}$ with this level, the following hold.*

(i) $|W^{\mathrm{hc}}(\mathcal{C})| \le \exp(B_E(\mathcal{C}) + I^{\mathrm{b}}_*(\mathcal{C}) + \Gamma_q(b_q(\mathcal{C}), \bar{g}))$.

(ii) *$W^{\mathrm{hc}}(\mathcal{C})$ does not depend on the order of the loops of $\mathcal{C}$; $W^{\mathrm{hc}}(h\mathcal{C}') = W^{\mathrm{hc}}(\mathcal{C}')$ for every translate $\mathcal{C}'$ of $\mathcal{C}$ and every map $h(x) = Ox+v$ of $\mathbb{R}^4$ with a signed permutation matrix O and $v \in \mathbb{R}^4$, with the convention* (1.6)*; $W^{\mathrm{hc}}(\bar{\mathcal{C}}) = \overline{W^{\mathrm{hc}}(\mathcal{C})}$; and $W^{\mathrm{hc}}(\mathcal{C})$ does not change if a loop C_i is replaced by C_i^{-1} and its character by $\bar{\chi}_i$.*

(iii) *With μ, y, θ, and $\mathcal{D}_1, \dots, \mathcal{D}_m$ as in Theorem 1.2(iii), the two matrices* (1.4) *with W^{hc} in place of W^∞ are Hermitian and positive semidefinite, and $W^{\mathrm{hc}}(\theta\bar{\mathcal{D}}_a) = \overline{W^{\mathrm{hc}}(\mathcal{D}_a)}$ for every a.*

With the convention (1.6) the reflection in (iii) may be taken in any hyperplane $\{x_\mu = y\}$, $y \in \mathbb{R}$, for the translates $\mathcal{D}_a + w$ by one vector w with $y - w_\mu \in uL^{-q}\mathbb{Z}$: writing θ_0 for the reflection which replaces x_μ by $2(y-w_\mu) - x_\mu$, we have $\theta(x+w) = \theta_0 x + w$, so that $(\mathcal{D}_a + w) \cup \theta(\bar{\mathcal{D}}_b + w) = (\mathcal{D}_a \cup \theta_0\bar{\mathcal{D}}_b) + w$, and the matrices of the translated family are those of the family $\mathcal{D}_1, \dots, \mathcal{D}_m$ for the reflection θ_0.

The convention extends from the translations to all isometries of $\mathbb{R}^4$, with one exception. An isometry g which is not a symmetry of the lattice maps a loop at level q to a path which is not a path of the lattice, and before we can speak of the invariance under g we must say what the function is on the image. The construction on the lattices $g(uL^{-K}\mathbb{Z}^4)$ gives the image the value of the original ensemble. This is consistent only if the limits agree on any two ensembles of the lattice which are images of each other under an isometry, and for most ensembles the only such isometries are symmetries of the lattice. We call a loop ensemble $\mathcal{C}$ at level q in $\mathbb{R}^4$ *exceptional* if there are two vectors e_α, e_β of the basis to which all the segments of its loops are parallel, and its loops do not all lie in one plane.

Corollary 1.4 (Euclidean Covariance)**.** *Let $(\mathcal{C}, q) \in \mathfrak{E}$, and let g be an isometry of $\mathbb{R}^4$ such that $g\mathcal{C}$ is a loop ensemble at level q in $\mathbb{R}^4$. If $\mathcal{C}$ is not exceptional, then $g\mathcal{C} = h\mathcal{C}$ for a map $h(x) = Ox + v$ with a signed permutation matrix O and $v \in uL^{-q}\mathbb{Z}^4$, and $W^{\mathrm{hc}}(g\mathcal{C}) = W^{\mathrm{hc}}(\mathcal{C})$. Consequently*

$$W^{\mathrm{hc}}(g\mathcal{C}) \coloneqq W^{\mathrm{hc}}(\mathcal{C}) \,, \tag{1.7}$$

the level being q, defines W^{hc} without ambiguity on the images under all isometries g of $\mathbb{R}^4$ of the ensembles $\mathcal{C}$ with $(\mathcal{C}, q) \in \mathfrak{E}$ which are not exceptional, and the function so defined is invariant under every isometry of $\mathbb{R}^4$.

The corollary contains no information on the limits beyond Corollary 1.3: it says that the lattice has no isometries between its ensembles other than its symmetries, so that the invariance under the others is a matter of moving the lattice with the ensemble. The exceptional ensembles are the first place where the invariance under a rotation is a statement about the limits. For a loop C at level q in a plane parallel to e_0 and e_1, and with $\varepsilon_q := uL^{-q}$, the ensembles $(C, C + 5\varepsilon_q e_2)$ and $(C, C + 3\varepsilon_q e_2 + 4\varepsilon_q e_3)$ are images of each other under a rotation which fixes the plane of C and is not a symmetry of the lattice, and we do not know whether W^{hc} takes the same value on them.

2 The Chessboard Estimate

The estimate rests on one Cauchy–Schwarz inequality for each reflection, and on an argument which shows that these inequalities, applied repeatedly, spread one function over all the blocks. We first set up the blocks and the maps which move a function from one block to another.

Let $s \geq 0$, $q \geq 0$, and $K \geq q$ be integers, let n be an integer with $-q \leq n \leq s$, and put $b := uL^n$ and $m := L^{s-n}$, so that $T_s = 2mb$ and $b \in uL^{-q}\mathbb{Z}$. Let $\Lambda := (\mathbb{Z}/2m\mathbb{Z})^4$. For $j \in \mathbb{Z}/2m\mathbb{Z}$ and $y \in \mathbb{R}/T_s\mathbb{Z}$ put $\psi_j(y) := y + jb$ if j is even and $\psi_j(y) := (j+1)b - y$ if j is odd; this is well defined, because $T_s = 2mb$ and the parity of j is defined modulo $2m$. For $t \in \Lambda$ let $\varphi_t(x) := (\psi_{t_0}(x_0), \dots, \psi_{t_3}(x_3))$ and $|t| := t_0 + \cdots + t_3$ modulo 2. The map φ_t is the composition of a translation by a vector of $b\mathbb{Z}^4 \subset uL^{-q}\mathbb{Z}^4$ with reflections θ_μ, so it belongs to the group $\mathfrak{G}_q$ of Paper, Section 1.3, and it maps the open block $B_0^\circ := (0, b)^4$ onto the open block $B_t^\circ := \prod_\mu (t_\mu b, (t_\mu + 1)b)$.

For $\mu \in \{0, \dots, 3\}$ and $k \in \mathbb{Z}$ the reflection $\theta_{\mu,kb}$ is a reflection of level q, since $kb \in uL^{-q}\mathbb{Z}$. For $t \in \Lambda$ let $\hat\theta_k t$ be obtained from t by replacing t_μ by $2k - 1 - t_\mu$. Then

$$\theta_{\mu,kb} \circ \varphi_t = \varphi_{\hat\theta_k t} \,, \qquad |\hat\theta_k t| \neq |t| \,. \tag{2.1}$$

Indeed, only the coordinate μ changes, and $\theta_{\mu,kb}$ replaces y by $2kb - y$: for even $j = t_\mu$ we get $2kb - jb - y = (2k-j)b - y = \psi_{2k-1-j}(y)$, since $2k - 1 - j$ is odd, and for odd j we get $2kb - (j+1)b + y = \psi_{2k-1-j}(y)$, since $2k - 1 - j$ is even.

The maps act on functions of the gauge field on $\mathbb{T}_K^{(s)}$ as in Paper, Section 12.1: $(F \circ h)(U) = F(hU)$ with $(hU)(b) = U(hb)$. Since $(h(h'U))(b) = (h'U)(hb) = U(h'hb)$, we have $h(h'U) = (h'h)U$ and

$$(F \circ h) \circ h' = F \circ (h'h) \,, \tag{2.2}$$

and if F depends only on the values of U on the bonds with both ends in a set X of sites, then $F \circ h$ depends only on those with both ends in hX. For a bounded measurable function F and $t \in \Lambda$ we put $\vartheta_t F := F \circ \varphi_t$ if $|t|$ is even and $\vartheta_t F := \overline{F \circ \varphi_t}$ if $|t|$ is odd. Since $(\Theta_{\mu,kb}F)(U) = \overline{F(\theta_{\mu,kb}U)}$, so that $\Theta_{\mu,kb}F = \overline{F \circ \theta_{\mu,kb}}$, the rules (2.2) and (2.1) give

$$\Theta_{\mu,kb}\vartheta_t F = \vartheta_{\hat\theta_k t} F \ : \tag{2.3}$$

for even $|t|$ the left side is $\overline{F \circ (\theta_{\mu,kb}\varphi_t)} = \overline{F \circ \varphi_{\hat\theta_k t}}$, with $|\hat\theta_k t|$ odd, and for odd $|t|$ it is $F \circ \varphi_{\hat\theta_k t}$, with $|\hat\theta_k t|$ even.

We first spread a function over the slabs of one direction. For $\mu \in \{0, \dots, 3\}$ and $j \in \mathbb{Z}/2m\mathbb{Z}$ we write $\vartheta_j^\mu := \vartheta_{je_\mu}$, which maps the open slab $\{0 < x_\mu < b\}$ onto the open slab $\{jb < x_\mu < (j+1)b\}$.

Lemma 2.1 (One Direction). *Let $\mu \in \{0,\dots,3\}$, and let G be a bounded measurable function of the gauge field on $\mathbb{T}_K^{(s)}$ which depends only on the values of U on the bonds with both ends in the open slab $\{0 < x_\mu < b\}$. Then, for every $S \subset \mathbb{Z}/2m\mathbb{Z}$,*

$$\mathbb{E}_K^{(s)}\Big[\prod_{j\in\mathbb{Z}/2m\mathbb{Z}} \vartheta_j^\mu G\Big] \ge 0 \qquad \textit{and} \qquad \Big|\mathbb{E}_K^{(s)}\Big[\prod_{j\in S} \vartheta_j^\mu G\Big]\Big|^{2m} \le \mathbb{E}_K^{(s)}\Big[\prod_{j\in\mathbb{Z}/2m\mathbb{Z}} \vartheta_j^\mu G\Big]^{|S|} . \tag{2.4}$$

In particular $|\mathbb{E}_K^{(s)}[G]|^{2m} \le \mathbb{E}_K^{(s)}[\prod_{j\in\mathbb{Z}/2m\mathbb{Z}} \vartheta_j^\mu G]$.

Proof. For $S \subset \mathbb{Z}/2m\mathbb{Z}$ put $Z(S) \coloneqq \mathbb{E}_K^{(s)}[\prod_{j\in S} \vartheta_j^\mu G]$, so that $Z(\emptyset) = 1$ and $Z(\{0\}) = \mathbb{E}_K^{(s)}[G]$. For $k \in \mathbb{Z}/2m\mathbb{Z}$ let $H_k^+ \coloneqq \{k, k+1, \dots, k+m-1\}$, let H_k^- be its complement, and let θ_k also denote the involution $j \mapsto 2k-1-j$ of $\mathbb{Z}/2m\mathbb{Z}$, which maps H_k^+ onto H_k^-.

We first show that, for every S and every k,

$$|Z(S)|^2 \le Z(S_k^+)\, Z(S_k^-) , \qquad S_k^\pm \coloneqq (S \cap H_k^\pm) \cup \theta_k(S \cap H_k^\pm) . \tag{2.5}$$

The open half-torus $H_{\mu,kb} = \{kb < x_\mu < kb + mb\}$ of the reflection $\theta_{\mu,kb}$ contains the slabs $\{jb < x_\mu < (j+1)b\}$ with $j \in H_k^+$, because $T_s/2 = mb$. The functions

$$A \coloneqq \prod_{j\in S\cap H_k^+} \vartheta_j^\mu G , \qquad B \coloneqq \prod_{j\in\theta_k(S\cap H_k^-)} \vartheta_j^\mu G \tag{2.6}$$

therefore depend only on the values of U on bonds with both ends in $H_{\mu,kb}$, which belong to the set E_+ of Paper, Section 12.1, so that A and B lie in $\mathcal{A}_+$. Since $\Theta_{\mu,kb}$ is multiplicative, (2.3) gives $\Theta_{\mu,kb}B = \prod_{j\in S\cap H_k^-} \vartheta_j^\mu G$ and $\Theta_{\mu,kb}A = \prod_{j\in\theta_k(S\cap H_k^+)} \vartheta_j^\mu G$, so that

$$Z(S) = \mathbb{E}_K^{(s)}[A\,\Theta_{\mu,kb}B] , \qquad Z(S_k^+) = \mathbb{E}_K^{(s)}[A\,\Theta_{\mu,kb}A] , \qquad Z(S_k^-) = \mathbb{E}_K^{(s)}[B\,\Theta_{\mu,kb}B] . \tag{2.7}$$

By Paper, Lemma 12.2, $\langle F, F'\rangle \coloneqq \mathbb{E}_K^{(s)}[F\,\Theta_{\mu,kb}F']$ is a positive semidefinite sesquilinear form on $\mathcal{A}_+$, and (2.5) is its Cauchy–Schwarz inequality $|\langle A, B\rangle|^2 \le \langle A, A\rangle\langle B, B\rangle$. In particular $Z(S_k^+) = \langle A, A\rangle \ge 0$ for every S, and for $S = \mathbb{Z}/2m\mathbb{Z}$, for which $S_k^+ = \mathbb{Z}/2m\mathbb{Z}$, this is the first assertion of (2.4).

For the second assertion we fix $\epsilon > 0$, put $\zeta \coloneqq Z(\mathbb{Z}/2m\mathbb{Z}) + \epsilon > 0$, and let $\rho(S) \coloneqq |Z(S)|\zeta^{-|S|/(2m)}$. Since $|S_k^+| + |S_k^-| = 2|S|$, the inequality (2.5) gives

$$\rho(S)^2 \le \rho(S_k^+)\,\rho(S_k^-) \tag{2.8}$$

for every S and every k. Let ρ_* be the largest of the finitely many numbers $\rho(S)$, so that $\rho_* \ge \rho(\emptyset) = 1$. If $\rho(S) = \rho_*$, then (2.8) and $\rho(S_k^\pm) \le \rho_*$ force $\rho(S_k^+) = \rho(S_k^-) = \rho_*$ for every k. Among the sets with $\rho(S) = \rho_*$ we choose one, S, of the largest cardinality. Then $|S_k^\pm| \le |S|$ for every k, and since the two cardinalities add up to $2|S|$, both are equal to $|S|$, that is, $|S\cap H_k^+| = |S|/2$ for every k. Comparing this for k and $k+1$ gives $\mathbf{1}_S(k+m) = \mathbf{1}_S(k)$, so that S is invariant under the shift $j \mapsto j+m$, which maps H_k^+ onto H_k^-. The set S_k^+ maximises ρ and has $|S|$ elements as well, so it is invariant under the same shift; since $S_k^+ \cap H_k^+ = S \cap H_k^+$ and $S_k^+ \cap H_k^- = \theta_k(S\cap H_k^+)$, this means that $(S\cap H_k^+)+m = \theta_k(S\cap H_k^+)$, while the invariance of S gives $(S\cap H_k^+)+m = S\cap H_k^-$. Hence $\theta_k(S\cap H_k^+) = S\cap H_k^-$, that is, $\theta_k S = S$, for every k. The involutions θ_k generate a group which acts transitively on $\mathbb{Z}/2m\mathbb{Z}$, because $\theta_{k+1}\theta_k$ is the shift by 2 and θ_0 maps 0 to -1; so S is either empty or all of $\mathbb{Z}/2m\mathbb{Z}$, and $\rho_* = \rho(S)$ is either 1 or $Z(\mathbb{Z}/2m\mathbb{Z})/\zeta < 1$. Since $\rho_* \ge 1$, we have $\rho_* = 1$, so that $\rho(S') \le 1$, that is, $|Z(S')| \le (Z(\mathbb{Z}/2m\mathbb{Z}) + \epsilon)^{|S'|/(2m)}$, for every $S' \subset \mathbb{Z}/2m\mathbb{Z}$. Letting $\epsilon \to 0$ gives the second assertion, and $S' = \{0\}$ gives the last one. *q.e.d.*

Applied in the four directions in turn, the lemma spreads a function of one block over all the blocks.

Lemma 2.2 (Chessboard Estimate)**.** *Let F be a bounded measurable function of the gauge field on $\mathbb{T}_K^{(s)}$ which depends only on the values of U on the bonds with both ends in B_0°. Then*

$$\mathbb{E}_K^{(s)}\Big[\prod_{t\in\Lambda}\vartheta_t F\Big] \geq 0 \qquad \textit{and} \qquad |\mathbb{E}_K^{(s)}[F]|^{|\Lambda|} \leq \mathbb{E}_K^{(s)}\Big[\prod_{t\in\Lambda}\vartheta_t F\Big]\,. \tag{2.9}$$

Proof. Put $G_0 := F$ and $G_{\nu+1} := \prod_{j\in\mathbb{Z}/2m\mathbb{Z}}\vartheta_j^\nu G_\nu$ for $\nu = 0,1,2,3$, and let Λ_ν be the set of the $t\in\Lambda$ with $t_{\nu'}=0$ for every $\nu'\geq\nu$. We claim that $G_\nu = \prod_{t\in\Lambda_\nu}\vartheta_t F$. This holds for $\nu=0$. If it holds for ν, then, since ϑ_j^ν is multiplicative and commutes with complex conjugation, it suffices to see that $\vartheta_j^\nu\vartheta_t F = \vartheta_{t+je_\nu}F$ for $t\in\Lambda_\nu$; and indeed $(F\circ\varphi_t)\circ\varphi_{je_\nu} = F\circ(\varphi_{je_\nu}\varphi_t)$ by (2.2), where $\varphi_{je_\nu}\varphi_t = \varphi_{t+je_\nu}$ because the two maps change different coordinates and ψ_0 is the identity, while the numbers of complex conjugations add up to $|t|+j = |t+je_\nu|$ modulo 2. The maps φ_t with $t\in\Lambda_\nu$ change only the coordinates $x_{\nu'}$ with $\nu'<\nu$ and map B_0° into $\{0<x_\nu<b\}$, so that G_ν depends only on the values of U on the bonds with both ends in this slab, and Lemma 2.1 with $\mu=\nu$ and $G=G_\nu$ gives $\mathbb{E}_K^{(s)}[G_{\nu+1}]\geq 0$ and $|\mathbb{E}_K^{(s)}[G_\nu]|^{2m}\leq\mathbb{E}_K^{(s)}[G_{\nu+1}]$. Since $G_4=\prod_{t\in\Lambda}\vartheta_t F$ and $|\Lambda|=(2m)^4$, the chain

$$|\mathbb{E}_K^{(s)}[F]|^{(2m)^4} \leq \mathbb{E}_K^{(s)}[G_1]^{(2m)^3} \leq \mathbb{E}_K^{(s)}[G_2]^{(2m)^2} \leq \mathbb{E}_K^{(s)}[G_3]^{2m} \leq \mathbb{E}_K^{(s)}[G_4] \tag{2.10}$$

proves (2.9). *q.e.d.*

The first step of this chain may be replaced by the stronger assertion of Lemma 2.1 for a set of slabs, which gives a bound for copies of F in some of the blocks of one row.

Corollary 2.3 (Blocks of One Row)**.** *Let F be as in Lemma 2.2, let $\mu\in\{0,\dots,3\}$, and let $S\subset\mathbb{Z}/2m\mathbb{Z}$. Then*

$$\Big|\mathbb{E}_K^{(s)}\Big[\prod_{j\in S}\vartheta_j^\mu F\Big]\Big|^{|\Lambda|} \leq \mathbb{E}_K^{(s)}\Big[\prod_{t\in\Lambda}\vartheta_t F\Big]^{|S|}\,. \tag{2.11}$$

Proof. The proof of Lemma 2.2 does not use the order in which the four directions are taken, and we take μ first. With $G_1' := \prod_{j\in\mathbb{Z}/2m\mathbb{Z}}\vartheta_j^\mu F$ it gives $\mathbb{E}_K^{(s)}[G_1']\geq 0$ and, by the last three steps of the chain (2.10), $\mathbb{E}_K^{(s)}[G_1']^{(2m)^3} \leq \mathbb{E}_K^{(s)}[\prod_{t\in\Lambda}\vartheta_t F]$. Lemma 2.1 with $G=F$ gives $|\mathbb{E}_K^{(s)}[\prod_{j\in S}\vartheta_j^\mu F]|^{2m}\leq\mathbb{E}_K^{(s)}[G_1']^{|S|}$, and raising this to the power $(2m)^3$ gives (2.11), because $|\Lambda|=(2m)^4$. *q.e.d.*

3 The Bound Uniform in the Volume

We now apply the chessboard estimate to a Wilson loop insertion. The reflected copies of an ensemble form an ensemble of the same kind, its normalisation is the product of the normalisations of the copies up to complex conjugation, and the uniform bound of the paper applies to it.

Proposition 3.1 (Bound Uniform in the Volume)**.** *Let $s\geq 0$, let $\mathcal{C}$ be a loop ensemble at level q on $\mathbb{T}^{(s)}$ whose loops are admissible, with $\bar g$ and q as in Paper, Theorem 1.5, for $\mathcal{C}$, let n be an integer with $-q\leq n\leq s$, and put $b := uL^n$. Assume that there is $v\in uL^{-q}\mathbb{Z}^4$ such that every site of $\mathcal{C}$ lies in the image in $\mathbb{T}^{(s)}$ of the open box $v+(0,b)^4$. Then*

$$|W^{(s)}(\mathcal{C})| \leq \exp\big(B_E(\mathcal{C}) + I_*^\flat(\mathcal{C}) + \Gamma_q(b,\bar g)\big)\,. \tag{3.1}$$

Proof. By Paper, Corollary 1.6(ii), which holds for $W_K^{(s)}$ at every $K > j_q$, the renormalised expectations do not change when $\mathcal{C}$ is translated by $-v \in uL^{-q}\mathbb{Z}^4$, and the translate has the same numbers B_E and $I_*^\flat$; we may therefore assume $v = 0$. We use the notation of Section 2 with these s, q, n, and $K > j_q$.

The reflected ensemble. For $t \in \Lambda$ let $\vartheta_t\mathcal{C}$ be the ensemble $\varphi_t\mathcal{C}$ of the images of the loops of $\mathcal{C}$ with the image orientations, with the characters of $\mathcal{C}$ if $|t|$ is even and with the conjugate characters if $|t|$ is odd. Since $\varphi_t \in \mathfrak{G}_q$, each $\vartheta_t\mathcal{C}$ is an ensemble at level q whose loops have the chord–arc constants, ratios P/r, and numbers P_q of those of $\mathcal{C}$, and $c_{\bar\pi} = c_\pi$ for every representation. The loops of $\vartheta_t\mathcal{C}$ lie in the open block B_t°, and these blocks are disjoint, so that the concatenation $\mathcal{D}$ of the ensembles $\vartheta_t\mathcal{C}$, $t \in \Lambda$, is a loop ensemble at level q with $|\Lambda|n$ admissible loops and $c_\mathcal{D} = c_\mathcal{C}$; hence $\bar g$ and q are as in Paper, Theorem 1.5, for $\mathcal{D}$ as well, the function $q_\sharp$ depending on the ensemble only through $c_\mathcal{D}$.

The chessboard estimate. The observable $F \coloneqq W_\mathcal{C}$ depends only on the values of U on the bonds of the loops of $\mathcal{C}$, which have both ends in the convex set B_0°. By Paper, Section 12.1, $W_\chi(C) \circ \varphi_t = W_\chi(\varphi_t C)$, and $\overline{W_\chi(C')} = W_{\bar\chi}(C')$ for every loop C', as in Paper, (12.5); by multiplicativity, $\vartheta_t W_\mathcal{C} = W_{\vartheta_t\mathcal{C}}$ and $\prod_{t\in\Lambda} \vartheta_t W_\mathcal{C} = W_\mathcal{D}$. Lemma 2.2 therefore gives

$$|\mathbb{E}_K^{(s)}[W_\mathcal{C}]|^{|\Lambda|} \le \mathbb{E}_K^{(s)}[W_\mathcal{D}] \,. \tag{3.2}$$

The normalisations. By Paper, Proposition 12.3, $\kappa_K^{(q)}(\varphi_t C_i, \chi) = \kappa_K^{(q)}(C_i, \chi)$ since $\varphi_t \in \mathfrak{G}_q$, and $\kappa_K^{(q)}(C_i, \bar\chi_i) = \overline{\kappa_K^{(q)}(C_i, \chi_i)}$. The normalisation of $\mathcal{D}$ is therefore the product over $t \in \Lambda$ of $\prod_i \kappa_K^{(q)}(C_i, \chi_i)$ or of its complex conjugate, and its modulus is $|\prod_i \kappa_K^{(q)}(C_i, \chi_i)|^{|\Lambda|}$, which is not zero because the step factors are not (Paper, Section 1.3). Dividing (3.2) by it gives

$$|W_K^{(s)}(\mathcal{C})|^{|\Lambda|} \le |W_K^{(s)}(\mathcal{D})| \,. \tag{3.3}$$

The uniform bound for $\mathcal{D}$. The requirements on $\bar g$ in Paper, Theorem 8.6, (8.19) and the hypothesis of Paper, Proposition 8.4, involve no representation and are among those which define γ_0, and the requirement (8.40) on γ_q is among those which define $q_\sharp$ (Paper, Section 1.3). That theorem gives, for $K \ge K_0(\mathcal{D})$,

$$|W_K^{(s)}(\mathcal{D})| \le e^{B_E(\mathcal{D}) + I_*(\mathcal{D})}(1 + \bar\delta_{\mathrm{LF}}) \,. \tag{3.4}$$

The number $B_E(\mathcal{D})$ of Paper, (7.32), is a sum over the loops of $\mathcal{D}$ of numbers which depend only on P/r, P_q, and $c_\mathcal{D} = c_\mathcal{C}$, so that $B_E(\mathcal{D}) = |\Lambda| B_E(\mathcal{C})$; and $I_*(\mathcal{D}) \le I_*^\flat(\mathcal{D}) = |\Lambda| I_*^\flat(\mathcal{C})$, because $P_{K'}(\mathcal{D}) = |\Lambda| P_{K'}(\mathcal{C})$. By Paper, Proposition 8.4, $\bar\delta_{\mathrm{LF}} = e^{2\bar C_K}(e^{\bar s_{\mathrm{LF}}} - 1)$ with $\bar C_K \coloneqq C_{\mathrm{fs}} N_q^4 (1 + \frac{1}{2}\log \hat g_{K'}^{-2})$ and $\bar s_{\mathrm{LF}} \coloneqq 2N_q^4 (MR_{K'})^{-4} \exp(-c_{\mathrm{LF}}\, p_0((1+\beta_0)\bar g))$, so that $1 + \bar\delta_{\mathrm{LF}} \le e^{2\bar C_K} e^{\bar s_{\mathrm{LF}}}$, because $e^{2\bar C_K} \ge 1$. Since $N_q = T_s L^{j_q}/u = 2mbL^{j_q}/u$ by Paper, (8.2), we have $N_q^4 = |\Lambda|(bL^{j_q}/u)^4$, and comparison with (1.2) gives $2\bar C_K + \bar s_{\mathrm{LF}} = |\Lambda| \Gamma_q(b, \bar g)$. Altogether,

$$|W_K^{(s)}(\mathcal{D})| \le \exp\Big(|\Lambda| (B_E(\mathcal{C}) + I_*^\flat(\mathcal{C}) + \Gamma_q(b, \bar g)) \Big) \,. \tag{3.5}$$

Conclusion. By (3.3) and (3.5), $|W_K^{(s)}(\mathcal{C})| \le \exp(B_E(\mathcal{C}) + I_*^\flat(\mathcal{C}) + \Gamma_q(b, \bar g))$ for every $K \ge \max(K_0(\mathcal{C}), K_0(\mathcal{D}))$, and the limit $K \to \infty$, which exists by Paper, Theorem 1.5, gives (3.1). *q.e.d.*

4 Proof of the Theorem

Proof of Theorem 1.2. The set $\mathfrak{E}$ is countable. For each q the loops at level q in $\mathbb{R}^4$ are finite paths in a countable lattice. The compact group G has countably many classes of irreducible

unitary representations, by the Peter–Weyl theorem, since $L^2(G)$ is separable and the matrix coefficients of inequivalent irreducible representations are orthogonal, and distinct classes have distinct characters. A countable union of sets of finite families of elements of countable sets is countable.

A bound uniform in s. Let $(\mathcal{C}, q) \in \mathfrak{E}$ and $s \geq s_q(\mathcal{C})$. Put $n := n_q(\mathcal{C})$, so that $-q \leq n \leq s-1$ and $b := uL^n = b_q(\mathcal{C})$, and let v_μ be the least coordinate x_μ of a site of $\mathcal{C}$ minus uL^{-q}, so that $v \in uL^{-q}\mathbb{Z}^4$. Every site of $\mathcal{C}$ lies in $v + [uL^{-q}, uL^{-q} + E(\mathcal{C})]^4 \subset v + (0,b)^4$, because $uL^{-q} + E(\mathcal{C}) < b$. The ensemble $\pi_s\mathcal{C}$ has admissible loops and $c_{\pi_s\mathcal{C}} = c_\mathcal{C}$, so that $\bar{g}$ and q are as in Paper, Theorem 1.5, for it, because γ_0 and $q_\sharp$ do not depend on the torus; and B_E and $I^\flat_*$ take the same values on $\pi_s\mathcal{C}$ and on $\mathcal{C}$. Proposition 3.1 gives

$$|W^{(s)}(\mathcal{C})| \leq \beta(\mathcal{C}) := \exp(B_E(\mathcal{C}) + I^\flat_*(\mathcal{C}) + \Gamma_q(b_q(\mathcal{C}), \bar{g})) \tag{4.1}$$

for every $s \geq s_q(\mathcal{C})$.

The subsequence. Let $e_1, e_2, \ldots$ be an enumeration of $\mathfrak{E}$, and let σ_i be the number $s_q(\mathcal{C})$ of $e_i = (\mathcal{C}, q)$. By (4.1) and the Bolzano–Weierstrass theorem we choose infinite sets $\mathbb{N} \supset N_1 \supset N_2 \supset \ldots$ with $N_i \subset [\sigma_i, \infty)$ along which $W^{(s)}(e_i)$ converges, and then integers $s_1 < s_2 < \ldots$ with $s_k \in N_k$. For every i the sequence $(s_k)_{k \geq i}$ lies in N_i, so that (1.3) exists for e_i, and (i) follows from (4.1).

The symmetries. Let $g(x) = Ox + v$ be as in (ii), and for every s let g_s be the map $x \mapsto Ox + v$ of $\mathbb{T}^{(s)}$, which belongs to the group $\mathfrak{G}_q$ of $\mathbb{T}^{(s)}$ and satisfies $\pi_s \circ g = g_s \circ \pi_s$, so that $\pi_s(g\mathcal{C}) = g_s\pi_s\mathcal{C}$. For $s \geq \max(s_q(\mathcal{C}), s_q(g\mathcal{C}))$, Paper, Corollary 1.6(ii), gives $W^{(s)}(g\mathcal{C}) = W^{(s)}(\mathcal{C})$, and the limit along (s_k) gives $W^\infty(g\mathcal{C}) = W^\infty(\mathcal{C})$. The other three statements of (ii) follow in the same way from Paper, Corollary 1.6(i), (iii), and (iv), the maps h and g_i being replaced by the maps of $\mathbb{T}^{(s)}$ with the same formulae, which are symmetries of the level q of $\mathbb{T}^{(s)}$.

Reflection positivity. Let μ, y, θ, and $\mathcal{D}_1, \ldots, \mathcal{D}_m$ be as in (iii). The ensembles $\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b$ are ensembles at level q in $\mathbb{R}^4$, since θ maps $uL^{-q}\mathbb{Z}^4$ onto itself and the loops of $\theta\bar{\mathcal{D}}_b$ lie in $\{x_\mu < y\}$; their loops are admissible, and $c_{\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b} \leq c$, since $c_{\bar{\pi}} = c_\pi$, so that they form pairs of $\mathfrak{E}$ with the level q, because $q_\sharp$ is nondecreasing in its first argument. Let s be so large that s exceeds the numbers s_q of all these ensembles and that $T_s/2 > x_\mu - y$ for every site x of every $\mathcal{D}_a$. Then π_s maps the sites of every $\mathcal{D}_a$ into the open half-torus $\{y < x_\mu < y + T_s/2\}$ of the reflection $\theta_{\mu,y}$ of $\mathbb{T}^{(s)}$, which is a reflection of level q, and $\pi_s \circ \theta = \theta_{\mu,y} \circ \pi_s$, so that $\pi_s(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) = \pi_s\mathcal{D}_a \cup \theta_{\mu,y}\overline{\pi_s\mathcal{D}_b}$. Paper, Corollary 1.7, applied on $\mathbb{T}^{(s)}$ to $\pi_s\mathcal{D}_1, \ldots, \pi_s\mathcal{D}_m$, says that the matrices (1.4) with $W^{(s)}$ in place of W^∞ are Hermitian and positive semidefinite, and that $W^{(s)}(\theta\bar{\mathcal{D}}_a) = \overline{W^{(s)}(\mathcal{D}_a)}$. These are closed conditions on finitely many numbers, and they pass to the limit along (s_k). *q.e.d.*

Proof of Corollary 1.3. (i) Each term of (1.5) satisfies the bound of Theorem 1.2(i) for $P_\sigma\mathcal{C}$, and B_E, $I^\flat_*$, and b_q take the same values on $P_\sigma\mathcal{C}$ as on $\mathcal{C}$, since they depend on the ensemble only through the ratios P/r, the numbers P_q, the number c, and the extent.

(ii) The first, third, and fourth statements follow term by term from Theorem 1.2(ii), since P_σ commutes with the reordering of the loops, with the conjugation of the characters, and with the reversal of the orientations. For the second, let $\mathcal{C}' = \mathcal{C} + w$. Then $h\mathcal{C}' = O\mathcal{C} + (Ow + v)$ is a translate of $O\mathcal{C}$, and $(O\mathcal{C}, q) \in \mathfrak{E}$, because O, like P_σ, maps $uL^{-q}\mathbb{Z}^4$ onto itself and preserves the distance $|\cdot|_\infty$, the segments, and the corners. By (1.6), $W^{\mathrm{hc}}(h\mathcal{C}') = W^{\mathrm{hc}}(O\mathcal{C})$ and $W^{\mathrm{hc}}(\mathcal{C}') = W^{\mathrm{hc}}(\mathcal{C})$, and it remains to show that $W^{\mathrm{hc}}(O\mathcal{C}) = W^{\mathrm{hc}}(\mathcal{C})$. We write $O = DP_\tau$ with a diagonal matrix D with entries ± 1 and a permutation τ, which is possible in exactly

one way. For every σ,

$$P_\sigma O = D_\sigma P_{\sigma\tau} \,, \qquad D_\sigma \coloneqq P_\sigma D P_\sigma^{-1} \,, \tag{4.2}$$

where D_σ is diagonal with entries ± 1, since $D_\sigma e_{\sigma(\mu)} = P_\sigma D e_\mu$ is $e_{\sigma(\mu)}$ or $-e_{\sigma(\mu)}$. Hence $P_\sigma(O\mathcal{C}) = D_\sigma(P_{\sigma\tau}\mathcal{C})$, and $W^\infty(P_\sigma O\mathcal{C}) = W^\infty(P_{\sigma\tau}\mathcal{C})$ by Theorem 1.2(ii) for the map $g = D_\sigma$. Summing over σ, and since $\sigma \mapsto \sigma\tau$ is a bijection of the permutations, $W^{\mathrm{hc}}(O\mathcal{C}) = W^{\mathrm{hc}}(\mathcal{C})$.

(iii) Let θ'_σ be the reflection of $\mathbb{R}^4$ which replaces $x_{\sigma(\mu)}$ by $2y - x_{\sigma(\mu)}$. Since $(P_\sigma \theta x)_{\sigma(\mu)} = 2y - x_\mu = 2y - (P_\sigma x)_{\sigma(\mu)}$ and the other coordinates of $P_\sigma\theta x$ and $P_\sigma x$ agree, $P_\sigma\theta = \theta'_\sigma P_\sigma$. Hence the loops of $P_\sigma\mathcal{D}_a$ lie in $\{x_{\sigma(\mu)} > y\}$ and $P_\sigma(\theta\bar{\mathcal{D}}_b) = \theta'_\sigma\overline{P_\sigma\mathcal{D}_b}$, so that

$$W^{\mathrm{hc}}(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) = \frac{1}{24}\sum_\sigma W^\infty(P_\sigma\mathcal{D}_a \cup \theta'_\sigma\overline{P_\sigma\mathcal{D}_b}) \,. \tag{4.3}$$

For each σ the matrix of the summands is Hermitian and positive semidefinite by Theorem 1.2(iii), applied to the direction $\sigma(\mu)$, the same y, and the ensembles $P_\sigma\mathcal{D}_1, \dots, P_\sigma\mathcal{D}_m$, which satisfy its hypotheses because P_σ preserves admissibility, the level, and the characters; so the average is Hermitian and positive semidefinite. The same theorem gives $W^\infty(\theta'_\sigma\overline{P_\sigma\mathcal{D}_a}) = \overline{W^\infty(P_\sigma\mathcal{D}_a)}$, and averaging gives $W^{\mathrm{hc}}(\theta\bar{\mathcal{D}}_a) = \overline{W^{\mathrm{hc}}(\mathcal{D}_a)}$. It remains to treat the connected matrix, in which the product of the averages is not the average of the products. We write $w_a^\sigma \coloneqq W^\infty(P_\sigma\mathcal{D}_a)$ and $w_a \coloneqq \frac{1}{24}\sum_\sigma w_a^\sigma = W^{\mathrm{hc}}(\mathcal{D}_a)$; expanding the last sum below and using $\frac{1}{24}\sum_\sigma w_a^\sigma = w_a$ gives

$$W^{\mathrm{hc}}(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) - w_a\overline{w_b} = \frac{1}{24}\sum_\sigma \Big[W^\infty(P_\sigma\mathcal{D}_a \cup \theta'_\sigma\overline{P_\sigma\mathcal{D}_b}) - w_a^\sigma\overline{w_b^\sigma}\Big] + \frac{1}{24}\sum_\sigma (w_a^\sigma - w_a)\overline{(w_b^\sigma - w_b)} \,. \tag{4.4}$$

The first sum is an average of matrices which are Hermitian and positive semidefinite by Theorem 1.2(iii), and the second is an average of the matrices $(z_a\bar{z}_b)_{a,b}$ with $z_a = w_a^\sigma - w_a$, which are Hermitian and positive semidefinite because $\sum_{a,b}\bar{\xi}_a z_a \bar{z}_b \xi_b = |\sum_a \bar{\xi}_a z_a|^2$; so the connected matrix of W^{hc} is Hermitian and positive semidefinite. *q.e.d.*

Remark 4.1 (What the Theorem Does Not Assert). The theorem gives no lower bound, and the lower bound of the paper does not pass to the limit: its proof, Paper, Proposition 12.5, uses $w_{\ell_0} \geq 1 - \bar{\delta}_{\mathrm{LF}}$ with a threshold $\bar{g}_1$ chosen so that $\bar{\delta}_{\mathrm{LF}} \leq 1/20$, and $\bar{\delta}_{\mathrm{LF}}$ grows with N_q, while a chessboard estimate bounds expectations from above only. The theorem therefore does not exclude that W^∞ vanishes on every nonempty ensemble, and Paper, Theorem 1.8, whose threshold $\bar{g}_0$ depends on T/u, has no counterpart here. The limits may depend on the subsequence.

5 The Dependence on the Separation

The limits of Theorem 1.2 are functions of ensembles in a lattice, and nothing said so far relates their values on two ensembles of which one is obtained from the other by moving some of its loops. Reflection positivity and the invariance under translations give such a relation for the simplest relative motion, in which two parts of an ensemble on the two sides of a coordinate hyperplane are moved apart along its normal. The mechanism is that of a transfer matrix: the translation along the normal is symmetric for the form of reflection positivity, the chessboard estimate bounds an ensemble and its mirror image uniformly in their distance, and a symmetric map whose powers stay bounded is a contraction, whose even powers decrease, and decrease ever more slowly. We carry this out on the matrices of Theorem 1.2(iii), without introducing a Hilbert space. The result is an estimate on each lattice, Theorem 5.3, and,

since the estimate holds at every level and the functions of two levels are proportional, the Lipschitz continuity of the limits in the separation, Theorem 5.4. Motions parallel to the hyperplane and deformations of the loops are not covered, and Remark 5.5 says what is missing.

Throughout this section $q \ge 0$ and $\mu \in \{0, \dots, 3\}$ are fixed, $\varepsilon_q := uL^{-q}$ is the spacing of the lattice of the level q, $y \in \varepsilon_q \mathbb{Z}$, and θ is the reflection of $\mathbb{R}^4$ which replaces x_μ by $2y - x_\mu$. For a loop ensemble $\mathcal{D}$ at level q in $\mathbb{R}^4$ and an integer r we write $\mathcal{D}_r := \mathcal{D} + r\varepsilon_q e_\mu$, and $\theta\bar{\mathcal{D}}_r$ for the mirror image $\theta\overline{\mathcal{D}_r}$ of $\mathcal{D}_r$.

Lemma 5.1 (Bounded Logarithmically Convex Sequences)**.** *Let $(a_r)_{r\ge 0}$ be nonnegative numbers with $a_r^2 \le a_{r-1}a_{r+1}$ for every $r \ge 1$ and $\liminf_{r\to\infty} a_r < \infty$. Then $a_{r+1} \le a_r$ for every $r \ge 0$.*

Proof. Every tail $(a_{r_0+r})_{r\ge 0}$ satisfies the hypotheses, so it suffices to show that $a_1 \le a_0$. Suppose that $a_1 > a_0$. Then $a_1 > 0$, and $a_1^2 \le a_0 a_2$ gives $a_0 > 0$. If a_{r-1} and a_r are positive and $a_r/a_{r-1} \ge a_1/a_0$, then $a_{r+1} \ge a_r^2/a_{r-1} \ge a_r a_1/a_0 > 0$; by induction $a_r \ge a_0(a_1/a_0)^r$ for every r, which tends to infinity and contradicts the hypothesis on the lower limit. *q.e.d.*

Proposition 5.2 (An Ensemble and Its Mirror Image)**.** *Let $(\mathcal{D}, q) \in \mathfrak{E}$, let the loops of $\mathcal{D}$ lie in $\{x_\mu > y\}$, and put $M_r := W^\infty(\mathcal{D}_r \cup \theta\bar{\mathcal{D}}_r)$ for integers $r \ge 0$. Then $M_r \ge 0$ and $M_{r+1} \le M_r$ for every $r \ge 0$, $M_r^2 \le M_{r-1}M_{r+1}$ for every $r \ge 1$, and, for every $r \ge 2b_q(\mathcal{D})/\varepsilon_q$,*

$$M_r \le \exp\Big(2(B_E(\mathcal{D}) + I_*^\flat(\mathcal{D}) + \Gamma_q(b_q(\mathcal{D}), \bar{g}))\Big) \,. \tag{5.1}$$

Proof. The ensembles $\mathcal{D}_r$, $r \ge 0$, are ensembles at level q whose loops are admissible and lie in $\{x_\mu > y\}$, with the characters of $\mathcal{D}$, so that Theorem 1.2(iii) applies to every finite family of them.

Positivity and convexity. Since $\theta(x + r'\varepsilon_q e_\mu) = \theta x - r'\varepsilon_q e_\mu$, the translation by $r'\varepsilon_q e_\mu$ maps $\theta\bar{\mathcal{D}}_{r'}$ to $\theta\bar{\mathcal{D}}$ and $\mathcal{D}_r$ to $\mathcal{D}_{r+r'}$, so that, by Theorem 1.2(ii),

$$W^\infty(\mathcal{D}_r \cup \theta\bar{\mathcal{D}}_{r'}) = W^\infty(\mathcal{D}_{r+r'} \cup \theta\bar{\mathcal{D}}) \tag{5.2}$$

for all $r, r' \ge 0$, a number which depends only on $r + r'$. Theorem 1.2(iii) for the single ensemble $\mathcal{D}_r$ gives $M_r \ge 0$. For the two ensembles $\mathcal{D}_{r-1}$ and $\mathcal{D}_{r+1}$, $r \ge 1$, it gives a positive semidefinite Hermitian matrix with the diagonal entries M_{r-1} and M_{r+1} and, by (5.2), with both other entries equal to $W^\infty(\mathcal{D}_{2r} \cup \theta\bar{\mathcal{D}}) = M_r$; its determinant is nonnegative, which is $M_r^2 \le M_{r-1}M_{r+1}$.

The bound for special r. Put $n := n_q(\mathcal{D})$ and $b := uL^n = b_q(\mathcal{D})$, so that $b \in \varepsilon_q\mathbb{Z}$ and $b \ge E(\mathcal{D}) + 2\varepsilon_q$, and let $d \ge \varepsilon_q$ be the least value of $x_\mu - y$ on the sites of $\mathcal{D}$. Let $j \ge 0$ be an even integer with $jb + \varepsilon_q \ge d$, and put $r := (jb + \varepsilon_q - d)/\varepsilon_q$, an integer which is not negative. Let $o \in \varepsilon_q\mathbb{Z}^4$ be the point with $o_\mu = y$ whose other coordinates o_ν are the least values of x_ν on the sites of $\mathcal{D}$, minus ε_q. By Theorem 1.2(ii), M_r does not change if $\mathcal{D}$ is translated by $-o$ and y is replaced by 0, and we assume $o = 0$. The sites of $\mathcal{D}_r$ then have $x_\mu \in [jb + \varepsilon_q, jb + \varepsilon_q + E(\mathcal{D})]$ and $x_\nu \in [\varepsilon_q, \varepsilon_q + E(\mathcal{D})]$ for $\nu \ne \mu$, and since $\varepsilon_q + E(\mathcal{D}) < b$, the sites of $\mathcal{D}^\flat := \mathcal{D}_r - jbe_\mu$ lie in $(0, b)^4$.

Let $s \ge s_q(\mathcal{D}_r \cup \theta\bar{\mathcal{D}}_r)$, so that $n \le s$, let $K > j_q$, and use the notation of Section 2 with these s, q, n, and K, writing the ensembles for their images under π_s. The function $F := W_{\mathcal{D}^\flat}$ depends only on the values of U on the bonds with both ends in B_0°, and $\vartheta_t F = W_{\vartheta_t\mathcal{D}^\flat}$ for every $t \in \Lambda$, as in the proof of Proposition 3.1. For $t = je_\mu$ the map φ_t is the translation by

jbe_μ, because j is even, so that $\vartheta_j^\mu F = W_{\mathcal{D}_r}$. For $t = (-1-j)e_\mu$ the map φ_t replaces x_μ by $\psi_{-1-j}(x_\mu) = -jb - x_\mu$, which is the composition of the translation by jbe_μ with θ, and $|t|$ is odd, so that $\vartheta^\mu_{-1-j} F = W_{\theta\bar{\mathcal{D}}_r}$. The two indices j and $-1-j$ are different modulo $2m$, their difference being odd, and Corollary 2.3 with $S = \{j, -1-j\}$ gives

$$|\mathbb{E}_K^{(s)}[W_{\mathcal{D}_r} W_{\theta\bar{\mathcal{D}}_r}]|^{|\Lambda|} \le \mathbb{E}_K^{(s)}[W_{\mathcal{D}^\sharp}]^2 \,, \tag{5.3}$$

where $\mathcal{D}^\sharp$ is the concatenation of the ensembles $\vartheta_t \mathcal{D}^\flat$, $t \in \Lambda$. Let κ be the product of the normalisations $\kappa_K^{(q)}$ of the loops of $\mathcal{D}$. By Paper, Proposition 12.3, the normalisation of $\mathcal{D}_r \cup \theta\bar{\mathcal{D}}_r$ is $\kappa\bar{\kappa}$, and the modulus of that of $\mathcal{D}^\sharp$ is $|\kappa|^{|\Lambda|}$, as in the proof of Proposition 3.1. Dividing (5.3) by $|\kappa|^{2|\Lambda|}$ and using the bound (3.5) for $\mathcal{D}^\sharp$, which was proved for the copies of every ensemble with sites in B_0° and in which B_E and $I_*^\flat$ take the same values on $\mathcal{D}^\flat$ as on $\mathcal{D}$, we obtain, for K large,

$$|W_K^{(s)}(\mathcal{D}_r \cup \theta\bar{\mathcal{D}}_r)|^{|\Lambda|} \le |W_K^{(s)}(\mathcal{D}^\sharp)|^2 \le \exp\Bigl(2|\Lambda|\bigl(B_E(\mathcal{D}) + I_*^\flat(\mathcal{D}) + \Gamma_q(b,\bar{g})\bigr)\Bigr) \,. \tag{5.4}$$

Taking the root, letting $K \to \infty$, and then $s \to \infty$ along (s_k), we obtain (5.1) for this r.

Monotonicity. The numbers r of the last step form an infinite set, so that $\liminf_r M_r < \infty$, and Lemma 5.1 gives $M_{r+1} \le M_r$ for every $r \ge 0$. The least even $j \ge 0$ with $jb \ge d - \varepsilon_q$ is less than $(d-\varepsilon_q)/b + 2$, so that the least of these numbers r is less than $2b/\varepsilon_q$, and (5.1) holds for every $r \ge 2b/\varepsilon_q$ by the monotonicity. *q.e.d.*

The bound (5.1) contains the cost Γ_q of the block of one copy, whatever the distance between the two copies, whereas Theorem 1.2(i) for the same ensemble contains the cost of a block which holds both. The monotonicity is the statement that the translation along e_μ is a contraction for the form of reflection positivity, and the next theorem is its consequence for two different ensembles.

Theorem 5.3 (Variation in the Separation)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be loop ensembles at level q in $\mathbb{R}^4$ whose loops are admissible, those of $\mathcal{A}$ in $\{x_\mu < y\}$ and those of $\mathcal{B}$ in $\{x_\mu > y\}$, and let $q \ge q_\sharp(c,\bar{g})$ for $c := \max(c_{\mathcal{A}}, c_{\mathcal{B}})$. For integers $r \ge 0$ put*

$$f(r) := W^\infty(\mathcal{A} \cup \mathcal{B}_r) \,, \qquad N := \bigl[W^\infty(\mathcal{A} \cup \theta\bar{\mathcal{A}})\, W^\infty(\mathcal{B} \cup \theta\bar{\mathcal{B}})\bigr]^{1/2} \,. \tag{5.5}$$

Then, for every $r \ge 0$,

$$|f(r)| \le N \,, \qquad |f(r+2) - f(r)| \le \frac{N}{\lfloor r/2 \rfloor + 1} \,, \qquad \sum_{k \ge 0} |f(r+2k+2) - f(r+2k)| \le N \,. \tag{5.6}$$

In particular $\lim_{k\to\infty} f(r+2k)$ exists. The same holds with W^{hc} in place of W^∞ in both definitions of (5.5).

Proof. Put $\mathcal{D}^1 := \mathcal{B}$ and $\mathcal{D}^2 := \theta\bar{\mathcal{A}}$. Both are ensembles at level q with admissible loops in $\{x_\mu > y\}$ and with $c_{\mathcal{D}^i} \le c$, and $\theta\bar{\mathcal{D}}^2 = \mathcal{A}$. For $i, i' \in \{1,2\}$ and integers $n \ge 0$ put $h_{ii'}(n) := W^\infty(\mathcal{D}^i_n \cup \theta\bar{\mathcal{D}}^{i'})$. As in (5.2), $W^\infty(\mathcal{D}^i_r \cup \theta\bar{\mathcal{D}}^{i'}_{r'}) = h_{ii'}(r+r')$ for all $r, r' \ge 0$, and $f(n) = h_{12}(n)$. By Theorem 1.2(iii), for every finite set R of nonnegative integers the matrix with the entries $h_{ii'}(r+r')$, indexed by the pairs (i,r) and (i',r') with $r, r' \in R$, is Hermitian and positive semidefinite; we write $\langle u, v\rangle_R := \sum u_{i,r} \bar{v}_{i',r'} h_{ii'}(r+r')$ for the corresponding form on the vectors $u = (u_{i,r})$.

Fix $\xi \in \mathbb{C}^2$, and for $k \ge 0$ put $a_k := \sum_{i,i'} \xi_i \bar{\xi}_{i'} h_{ii'}(2k)$. With $R = \{k\}$ and $u_{i,k} = \xi_i$ we get $a_k = \langle u, u\rangle_R \ge 0$. With $R = \{k-1, k+1\}$, $u_{i,k-1} = \xi_i$, $u_{i,k+1} = 0$, $v_{i,k-1} = 0$, and

$v_{i,k+1} = \xi_i$ we get $\langle u,u\rangle_R = a_{k-1}$, $\langle v,v\rangle_R = a_{k+1}$, and $\langle u,v\rangle_R = a_k$, so that the Cauchy–Schwarz inequality gives $a_k^2 \le a_{k-1}a_{k+1}$. With $R = \{k, k+2\}$ and $w_{i,k} = \xi_i$, $w_{i,k+2} = -\xi_i$ we get

$$a_k - 2a_{k+1} + a_{k+2} = \langle w,w\rangle_R \ge 0 \,. \tag{5.7}$$

The sequence is bounded: the matrix $(h_{ii'}(2k))_{i,i'}$ is positive semidefinite, so that $|h_{12}(2k)|^2 \le h_{11}(2k)h_{22}(2k)$, and $h_{ii}(2k) = W^\infty(\mathcal{D}_k^i \cup \theta\bar{\mathcal{D}}_k^i)$ is the number M_k of Proposition 5.2 for $\mathcal{D}^i$, which we denote by M_k^i and which is at most M_0^i; hence $a_k \le (|\xi_1|(M_0^1)^{1/2} + |\xi_2|(M_0^2)^{1/2})^2$. Lemma 5.1 therefore gives $a_{k+1} \le a_k$, and with (5.7) the differences $d_k := a_k - a_{k+1}$ satisfy $d_k \ge d_{k+1} \ge 0$. Consequently

$$(k+1)d_k \le \sum_{l=0}^{k} d_l = a_0 - a_{k+1} \le a_0 \,, \qquad \sum_{l\ge 0} d_l \le a_0 \,. \tag{5.8}$$

Since $d_k \ge 0$ for every ξ, the Hermitian matrix $(h_{ii'}(2k) - h_{ii'}(2k+2))_{i,i'}$ is positive semidefinite. Its diagonal entries are $\delta_k^i := M_k^i - M_{k+1}^i$, the numbers d_k of the two unit vectors ξ, and its other entries are $f(2k) - f(2k+2)$ and its conjugate, so that $|f(2k) - f(2k+2)|^2 \le \delta_k^1\delta_k^2$. By (5.8) for the unit vectors, $(k+1)\delta_k^i \le M_0^i$ and $\sum_l \delta_l^i \le M_0^i$, and $M_0^1 M_0^2 = N^2$ because $\mathcal{D}^2 \cup \theta\bar{\mathcal{D}}^2 = \theta\bar{\mathcal{A}} \cup \mathcal{A}$. Hence

$$|f(2k+2) - f(2k)| \le \frac{N}{k+1} \,, \qquad \sum_{l\ge 0} |f(2l+2) - f(2l)| \le \Big(\sum_l \delta_l^1\Big)^{1/2} \Big(\sum_l \delta_l^2\Big)^{1/2} \le N \,, \tag{5.9}$$

and $|f(2k)|^2 \le M_k^1 M_k^2 \le N^2$. This is (5.6) for even r. For odd r we apply the result to $\mathcal{B}_1$ in place of $\mathcal{B}$, for which f is replaced by $r \mapsto f(r+1)$ and N by $[W^\infty(\mathcal{A} \cup \theta\bar{\mathcal{A}})M_1^1]^{1/2} \le N$. The series in (5.6) is a tail of one of these two, and its convergence gives the limit.

For the last assertion, let f^σ and N^σ be the function and the number (5.5) of the ensembles $P_\sigma\mathcal{A}$ and $P_\sigma\mathcal{B}$, the direction $\sigma(\mu)$, and the same y, for a permutation σ. Then $P_\sigma(\mathcal{A} \cup \mathcal{B}_r) = P_\sigma\mathcal{A} \cup (P_\sigma\mathcal{B} + r\varepsilon_q e_{\sigma(\mu)})$, so that $W^{\mathrm{hc}}(\mathcal{A} \cup \mathcal{B}_r) = \frac{1}{24}\sum_\sigma f^\sigma(r)$, and each f^σ satisfies (5.6) with N^σ. By the Cauchy–Schwarz inequality and (4.3), $\frac{1}{24}\sum_\sigma N^\sigma \le [W^{\mathrm{hc}}(\mathcal{A} \cup \theta\bar{\mathcal{A}})\, W^{\mathrm{hc}}(\mathcal{B} \cup \theta\bar{\mathcal{B}})]^{1/2}$. *q.e.d.*

The theorem compares positions which differ by two steps of the lattice of the level q, at a distance of r steps, and its bound is the ratio of the two. The same ensembles are ensembles at every level $q' \ge q$, where the step is $\varepsilon_{q'} = uL^{-q'}$ and the ratio is smaller, but where the functions are normalised differently. The change of the normalisation is a factor which does not depend on the positions, and this turns the estimates of the single levels into a statement of continuity. We write $W_{q'}^\infty(\mathcal{C})$ for the limit (1.3) of the pair $(\mathcal{C}, q')$, and $\mathbb{D} := \bigcup_{q'\ge q} \varepsilon_{q'}\mathbb{Z}$, a dense subgroup of $\mathbb{R}$. Since L is odd, $2\mathbb{D} \ne \mathbb{D}$, and $\mathbb{D}$ is the union of the two dense sets $2\mathbb{D}$ and $\varepsilon_q + 2\mathbb{D}$.

Theorem 5.4 (Continuity in the Separation)**.** *Let $\mathcal{A}$, $\mathcal{B}$, c, and N be as in Theorem 5.3. There are numbers $c_{q'} \in \mathbb{C} \setminus \{0\}$, $q' \ge q$, with $c_q = 1$, and a function f on $\mathbb{D} \cap [0,\infty)$ such that*

$$W_{q'}^\infty(\mathcal{A} \cup (\mathcal{B} + te_\mu)) = c_{q'} f(t) \tag{5.10}$$

for every $q' \ge q$ and every $t \in \varepsilon_{q'}\mathbb{Z}$ with $t \ge 0$, and such that

$$|f(t') - f(t)| \le N\,\frac{t'-t}{t} \tag{5.11}$$

for all $t, t' \in \mathbb{D}$ with $0 < t < t'$ and $t' - t \in 2\mathbb{D}$. Consequently the restriction of f to each of the sets $2\mathbb{D} \cap (0,\infty)$ and $(\varepsilon_q + 2\mathbb{D}) \cap (0,\infty)$ extends to a continuous function on $(0,\infty)$, which satisfies (5.11) for all real $0 < t < t'$.

Proof. The constants. For $q' \geq q$ and $K > j_{q'}$ let $\nu_K^{q'}$ be the product of the normalisations $\kappa_K^{(q')}(C,\chi)$ of the loops of $\mathcal{A}$ and $\mathcal{B}$, which are loops at level q', and put $p_K^{q'} \coloneqq \nu_K^q / \nu_K^{q'}$. The stopping index is nondecreasing in the level, because $j_{q'} \geq q' + \bar{q}(j_{q'}) \geq q + \bar{q}(j_{q'})$ and j_q is the least integer with this property (Paper, Definition 3.13); by Paper, (1.5), $p_K^{q'}$ is therefore the product over the loops of the step factors $\kappa_k(C,\chi)$ with $K - j_{q'} < k \leq K - j_q$. By Paper, (7.23), and the bounds of Paper, Lemma 7.7, and Proposition 7.13, on its coefficients, for $k \geq 2$,

$$|\log \kappa_k(C,\chi)| \leq P_{k-1}(C)(C_{S'}\mathfrak{a}_k + C_E^R) + |\mathcal{K}(C)|(C_\zeta \mathfrak{a}_k + C_E^R) \,, \tag{5.12}$$

where $P_{k-1}(C) = \sum_v N_v^{(k-1)}(C)$ is the number of bonds of C on the lattice of the step $k-1$. For $k > K - j_{q'}$ this number is $P_q(C) L^{K-q-(k-1)} \leq P_q(C) L^{j_{q'}-q}$, by Paper, Lemma 3.14, and $\mathfrak{a}_k$ is bounded uniformly in K by Paper, (7.4). The numbers $\log p_K^{q'}$, sums of $j_{q'} - j_q$ terms (5.12) for each loop, are therefore bounded uniformly in K, and $p_K^{q'}$ stays in a compact subset of $\mathbb{C} \setminus \{0\}$. By a diagonal choice there are integers $K_1 < K_2 < \dots$ along which $p_K^{q'}$ converges for every $q' \geq q$, and we let $c_{q'} \neq 0$ be the limit; $c_q = 1$ because $p_K^q = 1$.

Two levels. Let $q \leq q' \leq q''$ and $t \in \varepsilon_{q'}\mathbb{Z}$, $t \geq 0$, and put $\mathcal{C}_t \coloneqq \mathcal{A} \cup (\mathcal{B} + t e_\mu)$, an ensemble at the levels q' and q'' with admissible loops. On a torus $\mathbb{T}^{(s)}$ with s large, the translation by te_μ belongs to the groups $\mathfrak{G}_{q'}$ and $\mathfrak{G}_{q''}$, so that the normalisations of $\mathcal{C}_t$ at the two levels are $\nu_K^{q'}$ and $\nu_K^{q''}$, by Paper, Proposition 12.3. Hence, for $K > j_{q''}$, the renormalised expectations at the two levels satisfy

$$W^{(s)}_{K,q''}(\mathcal{C}_t) = \frac{p_K^{q''}}{p_K^{q'}} \, W^{(s)}_{K,q'}(\mathcal{C}_t) \,. \tag{5.13}$$

Both renormalised expectations converge as $K \to \infty$, by Paper, Theorem 1.5; letting $K \to \infty$ along (K_i) and then $s \to \infty$ along (s_k) gives $W^\infty_{q''}(\mathcal{C}_t) = (c_{q''}/c_{q'}) W^\infty_{q'}(\mathcal{C}_t)$. The number $f(t) \coloneqq W^\infty_{q'}(\mathcal{C}_t)/c_{q'}$ therefore does not depend on the level q' with $t \in \varepsilon_{q'}\mathbb{Z}$, which is (5.10). In the same way, the normalisation of $\mathcal{A} \cup \theta\bar{\mathcal{A}}$ at the level q' is the product of that of $\mathcal{A}$ with its complex conjugate, by Paper, Proposition 12.3, and likewise for $\mathcal{B}$, so that the product of the renormalised expectations of $\mathcal{A} \cup \theta\bar{\mathcal{A}}$ and of $\mathcal{B} \cup \theta\bar{\mathcal{B}}$ at the level q' is $|p_K^{q'}|^2$ times that at the level q; in the limit, the number N of the level q' is $|c_{q'}|N$.

The estimate. Let $t, t' \in \mathbb{D}$ with $0 < t < t'$ and $t' - t \in 2\mathbb{D}$, and choose $q' \geq q$ with $t = r\varepsilon_{q'}$ and $t' = (r+2l)\varepsilon_{q'}$ for integers $r, l \geq 1$. Theorem 5.3 at the level q', whose hypotheses hold because $y \in \varepsilon_{q'}\mathbb{Z}$ and $q' \geq q_\sharp(c,\bar{g})$, gives after division by $|c_{q'}|$

$$|f(t') - f(t)| \leq \sum_{i=0}^{l-1} \frac{N}{\lfloor (r+2i)/2 \rfloor + 1} \leq \frac{lN}{\lfloor r/2 \rfloor + 1} \leq \frac{2lN}{r} = N \, \frac{t'-t}{t} \,. \tag{5.14}$$

On each of the two sets, which are dense in $(0,\infty)$ and any two points of which differ by an element of $2\mathbb{D}$, the function f is therefore Lipschitz continuous on $[t_0,\infty)$ with the constant N/t_0, for every $t_0 > 0$, and it has a unique continuous extension to $(0,\infty)$, to which (5.11) passes. *q.e.d.*

The constants $c_{q'}$ are the limits of the factors by which the normalisations of two levels differ, a finite renormalisation of the perimeter and corner terms as in Paper, (1.12). They are limits along the whole sequence of the cutoffs whenever the limit $W^{(s)}(\mathcal{A} \cup \mathcal{B}_r)$ of the level q is not zero for one torus and one r, since $p_K^{q'}$ is then the quotient of two convergent sequences by (5.13), the second of which has a limit different from zero; otherwise we know them only as limits along a subsequence.

Remark 5.5 (What Is Not Proved about the Positions). The two theorems concern one relative motion, along the normal of a coordinate hyperplane which separates the two parts, and three questions remain.

The two extensions of f in Theorem 5.4, from $2\mathbb{D}$ and from $\varepsilon_q + 2\mathbb{D}$, are not shown to agree. The estimates compare positions which differ by two steps of a lattice, because the translation by one step is symmetric for the form of reflection positivity but is not shown to be positive. What is missing is the reflection positivity of the lattice measures for the reflections in the hyperplanes midway between two neighbouring hyperplanes of sites, which Paper, Lemma 12.2, does not treat; with it the argument of this section applies to single steps.

For a motion of $\mathcal{B}$ parallel to the hyperplane, the translation preserves the form of reflection positivity instead of being symmetric for it, and positivity gives no estimate. The theorems apply to such a motion only if a second coordinate hyperplane, orthogonal to the motion, separates the two parts as well, and they do not apply to ensembles whose parts no coordinate hyperplane separates, such as two linked loops.

A deformation of a loop changes its normalisation. If it creates corners, the normalisation acquires the factor $\exp z_{v,w,K}$ of Paper, (7.34), for each of them, and the limits for the two loops are limits of expectations divided by different numbers, which nothing in the construction compares; perturbation theory attaches a divergent factor of its own to each corner [4, 2], so that we do not expect the limits to be continuous under such deformations for a normalisation fixed at one scale. A deformation which moves a segment of a loop parallel to itself keeps the corners. It is a motion of one part of a loop with respect to another, and the argument of this section would require the positivity for loops which cross the hyperplane and a bound on the limits which is exponential in the length of a loop, whereas $\Gamma_q(b_q(\mathcal{C}), \bar{g})$ grows like the fourth power of the extent. We have not carried this out.

References

[1] Tadeusz Bałaban, *Renormalization Group Approach to Lattice Gauge Field Theories. I. Generation of Effective Actions in a Small Field Approximation and a Coupling Constant Renormalization in Four Dimensions*, Comm. Math. Phys. **109** (1987), 249–301.

[2] Richard A. Brandt, Filippo Neri, and Masa-aki Sato, *Renormalization of Loop Functions for All Loops*, Phys. Rev. D **24** (1981), 879–902.

[3] Jürg Fröhlich, Robert Israel, Elliott H. Lieb, and Barry Simon, *Phase Transitions and Reflection Positivity. I. General Theory and Long Range Lattice Models*, Comm. Math. Phys. **62** (1978), 1–34.

[4] Alexander M. Polyakov, *Gauge Fields as Rings of Glue*, Nuclear Phys. B **164** (1980), 171–188.